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The effectiveness of two comprehensive recovery protocols on performance and physiological measures in elite soccer players: A parallel group-randomized trial

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Abstract

Elite soccer players consistently report using several recovery methods after practices and games. However, it is unclear how their subsequent performance could be enhanced using broad protocols. The aim of this study was to compare the effectiveness of two different comprehensive recovery protocols in physiological, neuromuscular, and perceptual outcomes. Eighteen Mexican National Team players (19.56 ± 0.62 years) were recruited. Using a randomized parallel group design, players followed one of two recovery protocols: (a) carbohydrate and protein shake, foam roller, cold-water immersion, and tart cherry juice concentrate ($n = 9$) or (b) carbohydrate and protein shake, stretching, and intermittent cold-water immersion ($n = 9$) following the completion of an unofficial game and the day after. Muscular creatine kinase, countermovement jump, hamstring maximal voluntary contraction, perceived recovery, and muscle soreness were assessed before, immediately after, and 20-44-68 h post-game. Significant effects (set at $\rho \leq 0.05$) of time were present in both interventions. Muscular creatine kinase was normalized entirely at 68 h post-game, while neuromuscular and perceptual outcomes were homogenized at 44 h. No statistically significant effects between protocols were found. Still, the interaction effects showed changes only in the group using protocol (a) at post-game and 20 h after in neuromuscular and muscle soreness. There were large correlations between muscular creatine kinase and accelerations, decelerations, sprints, and high-speed running distances. In conclusion, the interventions are equally effective for improving physiological, neuromuscular, and perceptual outcomes. Thus, elite soccer players may benefit from different combinations of methods after practices or games to obtain positive effects on recovery after them.

Keywords

Association football, carbohydrate and protein shake, cherry juice, cold-water immersion, creatine kinase, foam roller, global positioning system, high-speed running

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Introduction

Soccer is a complex, strenuous contact team sport characterized by unpredictable intermittent activity patterns interspersing low and high-intensity efforts with passive or active recovery.[1] From an organizational point of view, elite professional soccer is defined as a medium-density sport where the average player is exposed to high-congested fixture periods with a mean of 60 competitive games per season, 5.5 per month, 1–3 per week, or one every 4.3 days.[2] Frequent national team participation during the competitive season and travel issues may create an even more complex scenario to achieve a proper recovery before competitions.[3,4] This accumulation of events, games and

training sessions, combined with reduced recovery time between all these activities results in potential underperformance.[3] Thus, the ability to recover after competitive or intense training bouts is important to maintain or increase performance in subsequent efforts.[5]

Soccer-related physical performance is immediately impaired after a game and recovers to the previous levels gradually.[6] After a single game, total normalization of performance parameters such as sprint, jumping ability, or maximal strength is frequently observed after 72 h,[7] even with signs of biochemical and physiological stress.[8] However, when playing additional games during the week, subjective wellness values, perceived recovery, and physical performance parameters (e.g. sprint, jump, and strength) take longer, 96 h, to be normalized,[9,10] which increases the need for recovery strategies.[11] Moreover, some external load parameters of the game, such as high-intensity distance covered (total and per minute), high-speed running distance (HSR), and the number of sprints were significantly correlated to a creatine kinase (CK) increase, a muscle damage marker, at 24-h post-game.[12,13] These levels may remain increased for 3 days[14] and have to be considered as a basic marker that may be targeted by recovery strategies.[11] Therefore, quickly restoring pre-performance levels before the next competition is one of the main aims for soccer players.[15] Although protocols for specific soccer post-exercise recovery based on scientific evidence are available,[16] a gap exists between theory and practice in professional settings. The use of recovery interventions is not always based on research evidence,[17] which can negatively affect players' recovery and limit the ability to provide new guidelines to professional standards.[18] Thus, deep knowledge about physical performance profiles, time courses, players management, personal approaches, and players' awareness of recovery strategies post-practice/game can lead to faster performance recovery.[19]

Currently, scientific literature presents many methods to enhance recovery based on the type of practice, the time between sessions or competitions, and equipment or staff availability.[15] According to data from elite professional soccer staffs, the strategies most frequently used after a competition are an active recovery, stretching, food/fluid intake, ergogenic aids, cold/ice bath/shower/immersion, massage, and foam rolling.[17] However, to our knowledge, only three scientific studies present positive effects when combining different recovery methods in soccer.[20–22] Still, none of them have assessed the impact on health or performance, neither additive nor adverse effects in elite soccer players after a competition, although team staffs report combining more than one method when periodizing and performing recovery.[23] Therefore, the main aim of the current study was to compare the effectiveness of two comprehensive protocols on physiological, neuromuscular, and perceptual outcomes after an unofficial soccer game in elite players during a national team camp. It was hypothesized that a comprehensive recovery protocol combining more methods could speed up and be more effective in enhancing objective and subjective players' recovery parameters. In addition, as a secondary aim hypothesis, the acute changes of muscular CK (CK-III) correlate with the global positioning system (GPS) external load variables recorded during the game.

Methods

Experimental approach to the problem and participants

Twenty elite male soccer players (mean ± SD: age = 20 ± 1 years; height = 178.8 ± 5.0 cm; body mass = 72.9 ± 6.1 kg) from the under 21 (U-21) Mexican National Team were enrolled in this study. The study used a convenience sampling technique to select players from the U-21 Mexican National Team, allocating the players to the two groups using a block randomization technique.[24] All players had more than 10 years of experience playing soccer and performed 5–7 full-team soccer training sessions for a total training load of approximately 8–10 h per week, with at least one official game. Participants completed a medical questionnaire and initial screening. None of them had any diseases, and they did not smoke, drink alcohol, or take stimulants or substances that potentially alter hormonal responses. No participant presented any severe lower limb muscle injuries 2 months before the research. To ensure that the load and the intensity imposed were representative and similar, goalkeepers and outfield players who did not complete the total training camp sessions were excluded from the final analyses. After receiving information about the study aims, possible risks, benefits, and requirements, participants signed a written informed consent. Every participant could withdraw from the study at any moment. The study was conducted following the Declaration of Helsinki and its later amendments and was approved by the Research Ethics Committee of the University of Vic-Central University of Catalonia, Spain (registration number: 123/2020).

Design

An interventional balanced (1:1) parallel group-randomized trial was conducted to determine the effect of the two recovery protocol interventions after a 90-min unofficial soccer game. In this professional context, establishing a comparison group (control) was not suitable.[24] The study was conducted during the FIFA International break (August-September 2021) as a training camp developed for 9 days in Mexico DF. Before any data acquisition, the participants performed 6 days of similar practices. Food and fluid intake were controlled individually by a nutritionist using a food diary. Players were instructed to avoid any strenuous exercise

outside the training requirements nor modify usual sleeping patterns monitored and controlled, in quality[25] and duration. One familiarization session was developed to practice all the tests to ensure reliability, with a full explanation of the experimental protocols and testing procedures. Every participant had more than 1 year of experience using recovery methods and perceptual/well-being questionnaires. The outcomes were assessed on four different days (five time points). The first part of each testing session, except post-game, was designed to collect the players' perceptual outcomes. Immediately after that, the team physician took blood samples from all participants. Before performing the neuromuscular tests, a 5-min standardized dynamic warm-up using several different running patterns at an increasing intensity, joint mobility exercises, and dynamic stretches followed by two submaximal jumps was implemented (Figure 1). The testing battery lasted 25 min for every group of four players. Three days after the game, players participated in the regular training routine of the training camp.

The primary researcher was responsible for the stratified randomizing, using players' positions as the matching characteristic. Although being an unofficial game, licensed referees were present. Internal load data were collected after the game, using players' perceived exertion (RPE) rate on a 1–10 scale .[26,27] External data from the game were collected using valid and reliable WIMU® pro 10-Hz GPS devices (RealTrack Systems, Almería, Spain) which managed accelerometry data at 100 Hz.[28]

Before the start of the study, the protocol was registered at https://clinicaltrials.gov (ID: NCT04716049; https://clinicaltrials.gov/ct2/show/NCT04716049). Moreover, CONSORT guidelines were followed to report data (supplementary material I).

Outcomes

The same researchers supervised all the testing procedures and conducted them in an open official soccer facility at a temperature of 19 ± 1°C and humidity of 65.5 ± 6%.

Physiological outcomes. *Creatine kinase muscular isoenzyme III (CK-III or CK-MM).* Venous blood sample collection for CK-III isoform ratio in serum determination involved players passively drooling directly into a plastic tube. Samples were sealed and maintained at ideal conditions until analyses were carried out. CK-III was separated from total CK and measured in U/L using an electrophoretic method.[29]

Neuromuscular outcomes. *Hamstring maximal voluntary contraction (MVC).* Participants laid in a supine position, with the tested leg strapped around the ankle (in a neutral position), and performed a hip extension from 90° of hip

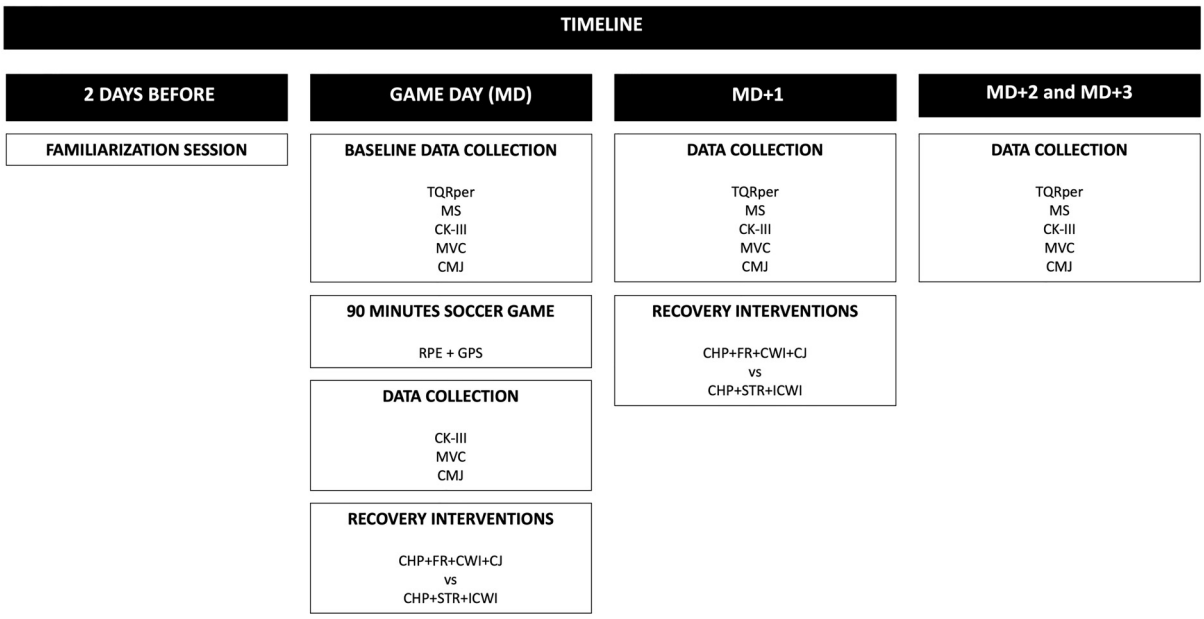

Figure 1. Protocol design timeline. CHP + FR + CWI + CJ: 250 mL of water with carbohydrate and protein, foam roller, cold-water immersion, and tart cherry juice concentrate; CHP + STR + ICWI: 250 mL of water with carbohydrate and protein, stretching, intermittent cold-water immersion; CK-III: creatine kinase muscular isoenzyme; MVC: hamstring maximal voluntary contraction; CMJ: countermovement jump; GPS: global positioning system; TQRper: total quality recovery perceived; MS: muscle soreness; MD + 1: 20 h after game; MD + 2: 44 h after game; MD + 3: 68 h after game; au: arbitrary units; RPE: rate of perceived exertion.

flexion (with a slight knee flexion of 20°). The nontested leg was resting on the floor and was fixed by one member of the research team. This test was adapted from Matinlauri and colleagues by using a strain gauge instead of a force plate.[30] Players placed their hands on the wall to avoid sliding on the floor. A strain gauge was placed between the ankle strap and the cord to evaluate the force (MVC, measured in Newtons) exerted at a sampling frequency of 80 Hz (Chronojump, Boscosystem®, Barcelona, Spain). The tester gave all participants the exact instructions; "to exert maximal force down to the floor" and continuous and standardized verbal motivation for 5 s. Each limb (dominant and non-dominant) was tested three times, in a randomized order and the average values were used in the subsequent analysis. The assessment of both legs was separated by 30 s, while the time separating two trials of the same leg was 60 s.[30]

Countermovement jump (CMJ) was performed using an Optojump® system (Microgate, Italy) with an accuracy of 1 kHz to determine the jump height (cm). For the CMJ, soccer players were required to bend their knees to approximately 90° angle and perform a maximal vertical thrust with the hands fixed on the hips.[31] Participants were instructed to keep their bodies upright throughout the jump and to land with their knees fully extended. Any jump that was perceived to deviate from the required instructions was repeated. Three trials were allowed, and the average values were used in subsequent analysis.[32] A 30-s passive recovery was provided between jumps.[33]

Perceptual outcomes. Total quality recovery perceived (TQRper). Players were asked to rate their recovery perception using the subjective validated TQRper questionnaire answering, "How recovered do you feel?". A scale ranging from 0 (very poorly recovered) to 10 (very well recovered) was used as a subjective measurement to assess fatigue.[34]

Muscle soreness (MS) was collected using a 5-point validated scale, with 5 being the best state and 0 being the worst.[25]

Procedures

After the soccer game, players were randomly assigned to one of two intervention groups: a carbohydrate and protein shake, foam roller, cold-water immersion, and tart cherry juice concentrate (CHP + FR + CWI + CJ) or a carbohydrate and protein shake, stretching, and intermittent cold-water immersion (CHP + STR + ICWI).

Both included a combination of natural and physical recovery methods and were performed twice, immediately after the game and the day after (20-h). All players drank a maximum of 500 mL of an isotonic sports drink and water "ad libitum" during the game.

"CHP + FR + CWI + CJ" intervention. Players drank 30 mL of CJ[35] and 250 mL of water with carbohydrates and protein within 30-min of completing the game; performed the foam rolling exercises using a polyvinylchloride pipe roller of 10.3-cm diameter and 0.3-cm thickness surrounded by a 1.5-cm thickness neoprene foam.[36] They were instructed and supervised to begin with the foam roller at the most proximal portion of the muscle and to roll as much body mass as tolerable from proximal to distal as smoothly as possible at a cadence of 1 s superior and 1 s inferior determined with a metronome. The foam roller activity was performed for 45 s, followed by a 15-s rest for each muscle group and in the same order (quadriceps, adductors, hamstrings, abductors, and gastrocnemius) in each lower extremity and repeated once. This protocol concluded with a cold-water immersion (CWI) where players immersed their lower body (up to the iliac crest) for 10 min in cold water (13–14°C)[37] inside a pool. Researchers closely monitored the water temperature every minute to maintain it.

"CHP + STR + ICWI" intervention. Players drank 250 mL of water with carbohydrates and protein within 30 min of completing the game. They performed 10 min of static stretching, involving two unilateral repetitions of 30 s, holding stretches to the quadriceps, adductors, hamstrings, abductors, and gastrocnemius muscles in each lower extremity.[38] An investigator supervised players during the stretching exercises. Intermittent cold-water immersion (ICWI) was the last recovery method used in this protocol. It consisted of five 2 min of intermittent immersions of the lower limb (up to the iliac crest) in a cold-water bath (13–14°C), separated by 2 min of rest in ambient air[39] using a fixed pool. Investigators closely monitored the water temperature every minute and monitored the 2-min immersions.

Statistical analyses

Data were presented as mean ± SD (or SE as stated) and 95% confidence intervals (CIs). Within-participant reliability of test measures (intra-session repeated efforts) was analyzed using the two-way random intraclass correlation coefficient (ICC) and coefficient of variation (CV). ICC values were >0.9 excellent, 0.9–0.75 good, 0.75–0.5 moderate, and <0.5 poor, and CV values were considered acceptable if <10%.

Independent samples *t*-test was used to check for baseline between-group differences in participants' characteristics and GPS outcomes. Cohen's *d* (mean difference/pooled SD) was reported for the analyses and interpreted as follows: <0.2 trivial; 0.2–0.5 small; 0.5–0.8 medium; >0.8 large.

The relationships among variables were analyzed using a linear mixed model (LMM) for continuous data and a generalized linear mixed model (GLMM) for categorical data. Both models considered the individuals as a random effect, and the time of testing (time) and the recovery strategy used

(group) as fixed effects. Mean and interaction effects were represented by the F for LMM and the chi-squared (χ^2) for GLMM. The GLMM used gamma distributions with identity as a link function. Estimated marginal means from LMM and GLMM with 95% CI were used to consider the effect of each factor. For those significant factors ($p <$ .05), a post hoc test with Holm adjustment was used to evaluate the interaction effect and the main effect of time or group. No missing data was present in the study. Still, two participants, one from each group, could not complete the study due to injuries that occurred during the game (see supplementary material II). Thus, 18 participants (CHP + FR + CWI + CJ = 9, CHP + STR + ICWI = 9) were finally included in the final analyses.

Pearson correlation coefficients were used to analyze the relationship between CK-III changes (pre- to post-game) and GPS game data. The magnitude of the relationships (r) was interpreted as follows: <0.1 trivial; 0.1–0.3 small; 0.3–0.5 medium; 0.5–0.7 large; 0.7–0.9 very large; and >0.9 extremely large.

Statistical analyses were conducted using the statistical package JASP for Macintosh (version 0.16.1, University of Amsterdam, Netherlands).

Results

Both groups had similar physical characteristics and game demands ($p > .05$; trivial to small effects, see supplementary material III). The reliability measures were for the CMJ (ICC = 0.90; 95% CI [0.82–0.95], CV = 4.6%; for the MVC (dominant) (ICC = 0.84; 95% CI [0.72–0.92], CV = 8.8%) and for the MVC (non-dominant) (ICC = 0.85; 95% CI [0.74–0.93], CV = 7.8%).

Main effects and interaction effects

The results from the LMM and GLMM are shown in Tables 1 and 2 and Figure 2. The main effect of time was large ($p < .001$, $F > 6.99$; and $\chi^2 > 46.49$, see Table 1) for all the outcomes. There was no main effect of the group (CHP + FR + CWI + CJ or CHP + STR + ICWI) for any of the outcomes ($p > .05$, see Table 1).

The interaction effect (group * time) was present for the CMJ, MVC (non-dominant limb), and TQRper models ($p < .05$, $F > 2.74$; and $\chi^2 = 10.58$, see Table 1,Figure 2).

Post hoc tests or contrasts

The effect of time for all the outcomes is presented in supplementary material IV and Figure 2.

The interaction effect for the CMJ and TQRper only revealed changes for the CHP + FR + CWI + CJ group at MD + 1 (supplementary material IV, Figure 2).

The interaction effect for the MVC (non-dominant) only revealed changes for the CHP + FR + CWI + CJ group at post-game and MD + 1 (supplementary material IV, Figure 2).

Relationship between CK-III and GPS data

The changes in CK-III from baseline to post-game related to the GPS data were performed by Pearson correlations and can be seen in Figure 3.

The changes in CK-III were correlated to the number of accelerations ($p < .001$; very large effect), the number of decelerations ($p = .003$; large effect), HSR distance ($p = .007$; large effect), and sprinting distance ($p = .04$; large effect). Instead, total distance ($p = .34$) was not correlated to the changes in CK-III.

Discussion

The present study was designed to compare the effectiveness of two recovery protocols after a game on physiological, neuromuscular, and perceptual outcomes with elite U-21 soccer players. Both protocols had a positive time effect, indicating that both could be beneficial at providing accelerated recovery after a game at different time points (20–44-68 h) depending on the variable of interest. However, no differences were found between them; therefore, this study cannot report the superiority of one protocol over the other. Based on this study, practitioners and players may choose different possibilities to ensure recovery effectiveness. Large and very large correlations between CK-III and external load parameters such as accelerations, decelerations, HSR distance, and sprinting distance were observed.

Top-tier soccer players show low levels of muscle damage during official tournaments and have faster recovery before the next game.[40] In this sense, current scientific evidence demonstrates that CK normalization time courses are up to ≥72 h, despite physical performance recovery lasting up to ≥48 h.[7] These findings are like those reported in the present study, showing that neuromuscular outcomes (MVC and CMJ) recovered faster than CK-III, returning to similar baseline levels at MD + 2. However, CK-III baseline levels were reached between 48 and 72 h,[41] where the peak CK-III levels were obtained at MD + 1 in both groups, like what has been reported in previous research.[42] CK-III isoform inter-subject variability response should be considered when using it as a recovery-specific marker.[43] Furthermore, four GPS parameters, such as HSR and sprint distance, accelerations, and decelerations, showed large to very large correlations with CK-III and muscle damage immediately after the game. While this was not one of the study's primary aims, the possibility of determining recovery needs using noninvasive markers coming from GPS technology could provide a useful framework to individualize the recovery process in different soccer settings.[12,15]

Table 1. Results from the linear mixed models (LMM) and the generalized linear mixed models (GLMM).

Linear mixed models	Variables		df	*F*	*d*
Interaction effects	CK-III		4, 64	0.20	0.94
	CMJ		4, 64	2.77	0.035
	MVC	Dominant	4, 64	1.07	0.38
		Non-dominant	4, 64	2.73	0.036
Time effect	CK-III		4, 64	35.54	<0.001
	CMJ		4, 64	6.99	<0.001
	MVC	Dominant	4, 64	10.01	<0.001
		Non-dominant	4, 64	8.91	<0.001
Group effect	CK-III		1, 16	0.31	0.586
	CMJ		1, 16	0.95	0.344
	MVC	Dominant	1, 16	1.67	0.214
		Non-dominant	1, 16	1.57	0.228
Generalized linear mixed models	Variables		df	χ^2	*p*
Interaction effects	TQRPer		3	10.58	.01
	MS		3	3.36	.34
Time effect	TQRPer		3	50.98	<.001
	MS		3	46.79	<.001
Group effect	TQRPer		1	0.11	.74
	MS		1	0.07	.96

CK-III: creatine kinase muscular isoenzyme; CMJ: countermovement jump; MVC: hamstring maximal voluntary contraction.
TQRper: total quality recovery perceived: MS: muscle soreness.

No decrement was found when analyzing the CMJ performance immediately after the game. Instead, different trends were observed depending on the group at MD + 1. Considering a real change in performance in any value above the CV (4.6% = 1.7 cm), the CHP + FR + CWI + CJ group showed a decrement of 3.2 cm, while the CHP + STR + ICWI group reported no changes (−1.1 cm). Although some differences can be observed in the groups, they are not statistically significant. In a recent investigation by Romagnoli and colleagues, where no recovery intervention was applied, CMJ was compromised at 48-h post-game compared to the baseline level,[44] suggesting the need to perform recovery strategies such as the ones used in this piece of research. Hence, applying recovery strategies seems important when lower body performance needs to be recovered within 48 h, with the CHP + STR + ICWI protocol providing slightly better benefits (although not significant compared to the other protocol) for maintaining CMJ performance following a game. Future studies must be implemented to confirm this claim.

When CJ was used for two consecutive days in combination with the other methods of the protocol (CHP + FR + CWI), this intervention did not significantly improve any variable analyzed compared to the CHP + STR + ICWI. Nevertheless, when using it as a unique recovery method for eight consecutive days and compared to a control group, neuromuscular outcomes (CMJ, MVC) recovered faster such as MS indicators were lower and inflammatory response was attenuated.[45] Although these beneficial effects, no effects on oxidative stress or CK were observed.[45] The intake frequency used in the protocol (only 2 days) and synergistic or additive possible effects when combining several recovery methods could attenuate CJ's impact.

The present study used two ways of applying immersion in cold water and other recovery methods. Both obtained similar results in terms of physiological, neuromuscular, and perceptual outcomes. CWI significantly affects delayed onset muscle soreness (DOMS), perceived fatigue and recovery, muscular power performance, and serum CK.[46] However, these effects were only observed 24 h after high-intensity exercise and not after eccentric loads. CWI was only effective for positively influencing muscular power but not muscular strength performance.[46] A moderate difference in favor of this recovery method (CWI) was observed in soccer at 48 h when analyzing CMJ and MVC.[47] Nevertheless, in a recent review, Schons and colleagues[48] found beneficial effects of CWI on muscle strength but no effects on jumping and running performance. Additionally, CWI and active recovery after a game did not improve jumping performance [22] in youth players. However, according to a recent study,[46] the duration and temperature of CWI can play a crucial role in enhancing recovery. For instance, lowering the time and temperature may improve the efficacy of CWI after high-intensity exercises, such as removing CK and incrementing endurance performance.

The literature lacks evidence supporting the muscular and physiological effects of stretching after physical activity or competitive soccer games on reducing MS.[49] Moreover, some authors concluded that it might even lead to an increase in DOMS and CK levels,[50] while others[49] found a CK reduction after performing static stretches.

Table 2. Descriptive of the outcome variables for group and time.

		CK-III				CMJ			
Time	Group	CK-III (U/L)	SE	95% CI		CMJ (cm)	SE	95% CI	
Baseline	CHP + FR + CWI + CJ	264.82	54.51	157.98	371.67	37.51	1.41	34.75	40.27
Post-game		394.84	54.51	287.99	501.68	36.49	1.41	33.73	39.24
MD + 1		424.48	54.51	317.63	531.32	34.29	1.41	31.53	37.05
MD + 2		311.18	54.51	204.34	418.02	36.54	1.41	33.79	39.3
MD + 3		248.38	54.51	141.53	355.22	38.44	1.41	35.68	41.19
Baseline	CHP + STR + ICWI	220.56	54.51	113.71	327.4	38.86	1.41	36.1	41.62
Post-game		360.56	54.51	253.72	467.41	38.43	1.41	35.67	41.18
MD + 1		368.27	54.51	261.43	475.11	37.72	1.41	34.96	40.47
MD + 2		267.83	54.51	160.99	374.68	38.85	1.41	36.09	41.61
MD + 3		221.61	54.51	114.77	328.46	38.57	1.41	35.81	41.33
		MVC (dominant)				**MVC (non-dominant)**			
Time		**MVC (*N*)**	**SE**	**95% CI**		**MVC (*N*)**	**SE**	**95% CI**	
Baseline	CHP + FR + CWI + CJ	361.86	22.82	317.13	406.59	357.3	22.2	313.8	400.8
Post-game		311.65	22.82	266.93	356.38	289.2	22.2	245.6	332.7
MD + 1		349.01	22.82	304.28	393.74	321.1	22.2	277.6	364.7
MD + 2		353.54	22.82	308.81	398.27	338.8	22.2	295.3	382.3
MD + 3		376.95	22.82	332.22	421.68	357.0	22.2	313.5	400.6
Baseline	CHP + STR + ICWI	406.39	22.82	361.66	451.12	379.1	22.2	335.6	422.6
Post-game		371.01	22.82	326.28	415.74	361.1	22.2	317.6	404.6
MD + 1		375.14	22.82	330.41	419.87	361.1	22.2	317.6	404.7
MD + 2		386.05	22.82	341.32	430.78	362.5	22.2	319.0	406.0
MD + 3		409.61	22.82	364.89	454.34	383.3	22.2	339.8	426.8
		TQRper				**MS**			
Time	**Group**	**TQRPer (au)**	**SE**	**95% CI**		**MS (au)**	**SE**	**95% CI**	
Baseline	CHP + FR + CWI + CJ	7.56	0.67	6.24	8.87	4	0.34	3.34	4.66
MD + 1		4.14	0.54	3.08	5.2	2.71	0.3	2.12	3.3
MD + 2		6.34	0.62	5.12	7.55	3.36	0.32	2.74	3.98
MD + 3		8.61	0.71	7.21	10	3.79	0.33	3.15	4.44
Baseline	CHP + STR + ICWI	6.93	0.62	5.72	8.15	3.66	0.33	3.02	4.3
MD + 1		5.69	0.57	4.57	6.8	2.56	0.3	1.98	3.14
MD + 2		6.8	0.61	5.6	8	3.52	0.32	2.88	4.15
MD + 3		8.17	0.67	6.85	9.48	4.05	0.34	3.39	4.72

CHP+FR+CWI+CJ: 250 mL of water with carbohydrate and protein, foam roller, cold-water immersion, and tart cherry juice concentrate; CHP+STR +ICWI: 250 mL of water with carbohydrate and protein, stretching, intermittent cold-water immersion; CK-III: creatine kinase muscular isoenzyme; CMJ: countermovement jump; MVC: hamstring maximal voluntary contraction; MS: muscle soreness; TQRPer: total quality recovery perceived; MD+1: 20 hours after game; MD+2: 44 hours after game; MD+3: 68 hours after game.

Therefore, the existing evidence is quite conflicting. However, when stretching exercises were combined with other recovery techniques, such as massage (after a competition),[51] it induced a lower perception of leg soreness and better CMJ values.[52] Previous research has found that CMJ values were significantly higher 24 h after a training session when using a combination of recovery methods consisting of submaximal running and static stretching.[21] These findings are similar to the ones in the present study, where stretching within a recovery protocol induced positive changes in the variables studied; yet, it is impossible to state that these improvements were caused by stretching or other recovery strategies.

A recent review has reported ambiguous conclusions and recommendations about foam rolling when analyzing the use among different-level athletes.[53] In another recent study, after a typical soccer session, foam rolling had a large effect on the recovery of agility, TQRper, and perceived MS compared to a passive recovery. Nevertheless, and according to the current study findings, similar perceptual and neuromuscular outcomes were found when comparing two protocols that combined different recovery methods. Nonetheless, and due to these non-conclusive results, foam rolling may be recommended as a recovery tool for this typology of athletes.

Despite the lack of definitive scientific evidence, some recovery methods, such as stretching or foam rolling, are widely used by elite soccer teams after training and competition.[17] One of the reasons could be that these recovery methods can influence players' perceptions and may increase adherence to the protocols.[21] Therefore, practitioners should consider their use when planning and

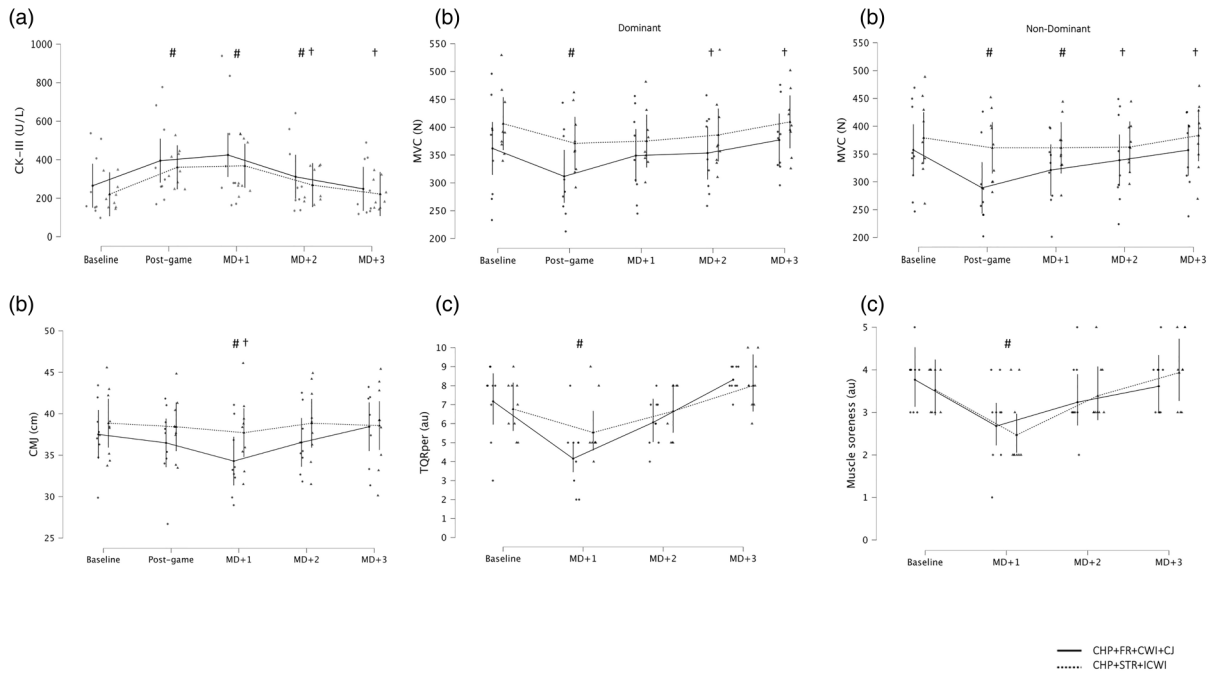

Figure 2. Physiological (a), neuromuscular (b), and perceptual (c) outcomes. CHP + FR + CWI + CJ: 250 mL of water with carbohydrate and protein, foam roller, cold-water immersion, and tart cherry juice concentrate; CHP + STR + ICWI: 250 mL of water with carbohydrate and protein, stretching, intermittent cold-water immersion; CK-III: creatine kinase muscular isoenzyme; MVC: hamstring maximal voluntary contraction; CMJ: countermovement jump; TQRper: total quality recovery perceived; MS: muscle soreness; MD + 1: 20 h after game; MD + 2: 44 h after game; MD + 3: 68 h after game; au: arbitrary units. Notes: # different from baseline. † different from post-game.

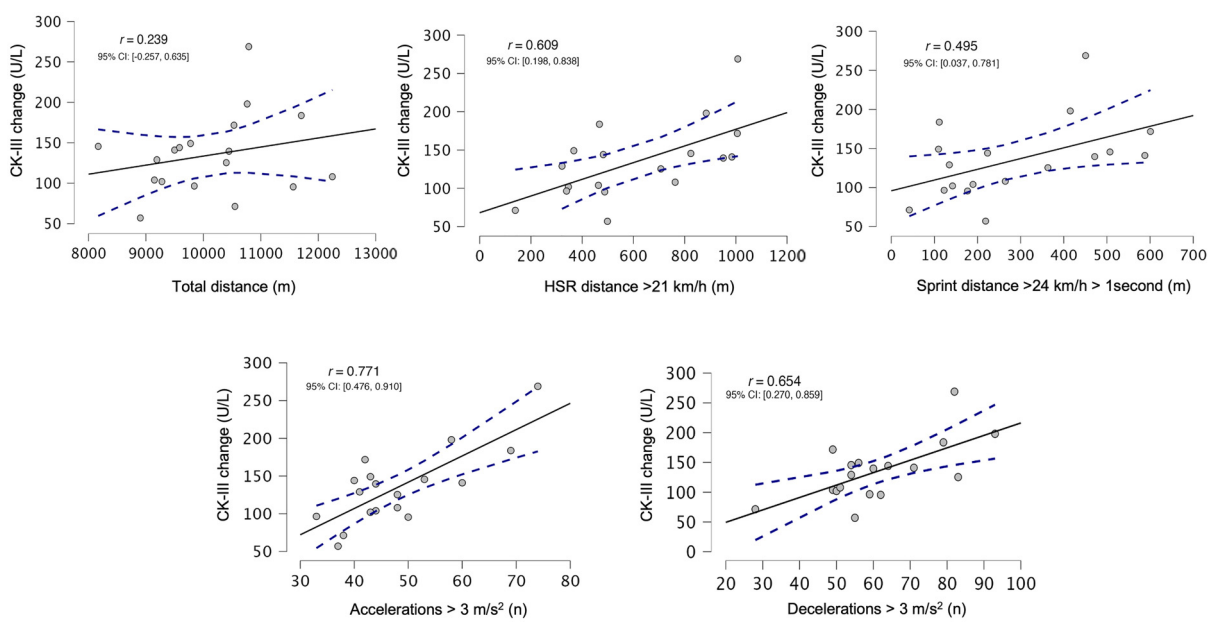

Figure 3. Relationship between CK-III and GPS variables. CK-III: creatine kinase muscular isoenzyme; HSR: High-Speed Running distance.

designing multimethod recovery protocols. Based on this study's findings, we recommend including different recovery methods based on several criteria, such as scientific agreement, but first, considering players' preferences and perceptions based on wellness, especially at elite levels.

This study found a time effect on recovery parameters in both interventions. Specifically, CK-III was normalized at MD + 3 and neuromuscular and perceptual outcomes at MD + 2. However, no differences between protocols were found, except for the interaction effect found in the CHP + FR + CWI + CJ group at post-game and MD + 1 in neuromuscular and TQRper outcomes. Therefore, based on these results, this paper's authors consider both protocols similarly effective. Practitioners could combine the recovery methods proposed in this study as post-competition recovery strategies and adjust the interventions to meet elite soccer players' requests and team contexts based on their fixtures or training schedule.

The protocol used a convenience sampling technique to select players from the U-21 Mexican National Team. Consequently, the sample size ($N = 20$) is relatively small, which could have limited the statistical power and affected the significance of some comparisons (between effects).[24] However, using "sample quality criteria," the specificity of the players enrolled (elite young soccer players) may have increased its representativity. Moreover, the players in the study were only male; therefore, these protocols should be replicated in female populations of a similar competitive level before their implementation. Finally, the lack of a control group in the study limits the possibility of concluding our investigation by understanding if any experimental conditions were more effective than controls in a similar population. However, this is a common limitation when professional players are enrolled.

Practical applications

- Elite soccer players could combine different recovery methods after games, such as the two combinations used in this study that reported a positive effect on recovery.
- Both protocols used in this study had a positive time effect, but since no differences were found between them. Practitioners could use the ones more suitable for their players.
- It is suggested that practitioners and players be given a choice among different options based on their preferences and the perceived effectiveness of recovery.
- Since large and very large correlations between CK-III and external load parameters such as accelerations, decelerations, HSR, and sprint distance were observed. Practitioners are invited to monitor these parameters to manage the recovery strategies better.

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Supplemental material

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References

1. Clemente FM, Rabbani A, Conte D, et al. Training/match external load ratios in professional soccer players: a full-season study. *Int J Environ Res Public Health* 2019; 16: 3057.
2. Anderson L, Orme P, di Michele R, et al. Quantification of training load during one-, two- and three-game week schedules in professional soccer players from the English premier league: implications for carbohydrate periodisation. *J Sports Sci* 2016; 34: 1250–1259.
3. Nédélec M, McCall A, Carling C, et al. Recovery in soccer: part I-post-match fatigue and time course of recovery. *Sports Med* 2012; 42: 997–1015.
4. Oliveira C, Ferreira D, Caetano C, et al. Nutrition and supplementation in soccer. *Sports* 2017; 5: 28.
5. Rey E, Lago-Peñas C, Lago-Ballesteros J, et al. The effect of recovery strategies on Contractile properties using tensiomyography and perceived muscle soreness in professional soccer players. *J Strength Cond Res* 2012; 26: 3081–3088.
6. Dupont G, Nédélec M, McCall A, et al. Football recovery strategies. Practical aspects of blending science and reality. *Aspetar Sport Medicine Journal* 2015; 4: 20–27.
7. Doeven SH, Brink MS, Kosse SJ, et al. Postmatch recovery of physical performance and biochemical markers in team ball sports: a systematic review. *BMJ Open Sport Exerc Med* 2018; 4: e000264.
8. Marqués-Jiménez D, Calleja-González J, Arratibel-Imaz I, et al. Biochemical and physical performance responses to a soccer match after a 72-hour recovery period. *Sports* 2022; 10: 140.
9. Silva JR, Rumpf MC, Hertzog M, et al. Acute and residual soccer match-related fatigue: a systematic review and meta-analysis. *Sports Med* 2018; 48: 539–583.
10. Carling C, Lacome M, McCall A, et al. Monitoring of post-match fatigue in professional soccer: welcome to the real world. *Sports Med* 2018; 48: 2695–2702.
11. Haller N, Hübler E, Stöggl T, et al. Evidence-based recovery in soccer – low-effort approaches for practitioners. *J Hum Kinet* 2022; 82: 75–99.
12. da Silva CD, Machado G, Fernandes AA, et al. Muscle damage-based recovery strategies can be supported by

predictive capacity of specific global positioning system accelerometry parameters immediately a post-soccer match-load. *J Strength Cond Res* 2021; 35: 1410–1418.
13. Russell M, Sparkes W, Northeast J, et al. Relationships between match activities and peak power output and Creatine Kinase responses to professional reserve team soccer match-play. *Hum Mov Sci* 2016; 45: 96–101.
14. Ispirlidis I, Fatouros IG, Jamurtas AZ, et al. Time-course of changes in inflammatory and performance responses following a soccer game. *Clin J Sport Med* 2008; 18: 423–431.
15. Calleja-González J, Mielgo-Ayuso J, Sampaio J, et al. Brief ideas about evidence-based recovery in team sports. *J Exerc Rehabil* 2018; 14: 545–550.
16. Abaïdia A-E and Dupont G. Recovery strategies for football players. *Swiss Sports Exerc Med* 2018; 66: 28–36.
17. Altarriba-Bartes A, Peña J, Vicens-Bordas J, et al. The use of recovery strategies by Spanish first division soccer teams: a cross-sectional survey. *Phys Sportsmed* 2021; 49: 297–307.
18. Field A, Harper LD, Chrismas BCR, et al. The use of recovery strategies in professional soccer: a worldwide survey. *Int J Sports Physiol Perform* 2021; 16: 1804–1815.
19. Rowell AE, Aughey RJ, Hopkins WG, et al. Identification of sensitive measures of recovery after external load from football match play. *Int J Sports Physiol Perform* 2017; 12: 969–976.
20. García-Concepción MA, Peinado AB, Paredes Hernández V, et al. Efficacy of different recovery strategies in elite football players. *Revista Internacional de Medicina y Ciencias de la Actividad Física y del Deporte* 2015; 15: 355–389.
21. Rey E, Lago-Peñas C, Casáis L, et al. The effect of immediate post-training active and passive recovery interventions on anaerobic performance and lower limb flexibility in professional soccer players. *J Hum Kinet* 2012; 31: 121–129.
22. Kinugasa T and Kilding AE. A comparison of post-match recovery strategies in youth soccer players. *J Strength Cond Res* 2009; 23: 1402–1407.
23. Querido SM, Brito J, Figueiredo P, et al. Postmatch recovery practices carried out in professional football: a survey of 56 Portuguese professional football teams. *Int J Sports Physiol Perform* 2022; 17: 748–754.
24. Beato M. Recommendations for the design of randomized controlled trials in strength and conditioning. Common design and data interpretation. *Front Sports Act Living* 2022; 4: 981836.
25. McLean BD, Coutts AJ, Kelly V, et al. Neuromuscular, endocrine, and perceptual fatigue responses during different length between-match microcycles in professional rugby league players. *Int J Sports Physiol Perform* 2010; 5: 367–383.
26. Impellizzeri FM, Rampinini E, Coutts AJ, et al. Use of RPE-based training load in soccer. *Med Sci Sports Exerc* 2004; 36: 1042–1047.
27. Borg GA. Psychophysical bases of perceived exertion. *Med Sci Sports Exerc* 1982; 14: 377–381.
28. Bastida Castillo A, Gómez Carmona CD, de la Cruz Sánchez E, et al. Accuracy, intra- and inter-unit reliability, and comparison between GPS and UWB-based position-tracking systems used for time-motion analyses in soccer. *Eur J Sport Sci* 2018; 18: 450–457.
29. Uji Y, Karmen A, Sugiuchi H, et al. Simultaneous automated measurement of serum total CK and CK-MM lsoform ratio in serum. *J Clin Lab Anal* 1994; 8: 273–276.
30. Matinlauri A, Alcaraz PE, Freitas TT, et al. A comparison of the isometric force fatigue-recovery profile in two posterior chain lower limb tests following simulated soccer competition. *PLoS One* 2019; 14: e0206561.
31. Rodacki ALF, Fowler NE and Bennett SJ. Vertical jump coordination: fatigue effects. *Med Sci Sports Exerc* 2002; 34: 105–116.
32. Claudino JG, Cronin J, Mezêncio B, et al. The countermovement jump to monitor neuromuscular status: a meta-analysis. *J Sci Med Sport* 2017; 20: 397–402.
33. Riggs MP and Sheppard JM. The relative importance of strength and power qualities to vertical jump height of elite beach volleyball players during the counter-movement and squat jump. *J Hum Sport Exerc* 2009; 4: 221–236.
34. Laurent CM, Green JM, Bishop PA, et al. A practical approach to monitoring recovery: development of a perceived recovery status scale. *J Strength Cond Res* 2011; 25: 620–628.
35. Doma K, Gahreman D and Connor J. Fruit supplementation reduces indices of exercise-induced muscle damage: a systematic review and meta-analysis. *Eur J Sport Sci* 2021; 21: 562–579.
36. Wiewelhove T, Döweling A, Schneider C, et al. A meta-analysis of the effects of foam rolling on performance and recovery. *Front Physiol* 2019; 10: 376.
37. Hohenauer E, Costello JT, Stoop R, et al. Cold-water or partial-body cryotherapy? Comparison of physiological responses and recovery following muscle damage. *Scand J Med Sci Sports* 2018; 28: 1252–1262.
38. Ryan ED, Herda TJ, Costa PB, et al. Determining the minimum number of passive stretches necessary to alter musculotendinous stiffness. *J Sports Sci* 2009; 27: 957–961.
39. Delextrat A, Calleja-González J, Hippocrate A, et al. Effects of sports massage and intermittent cold-water immersion on recovery from matches by basketball players. *J Sports Sci* 2013; 31: 11–19.
40. Bok D and Jukić I. Muscle damage during a soccer world cup preparatory and competition period. *Int J Sports Physiol Perform* 2020; 15: 496–502.
41. Brancaccio P, Limongelli FM and Maffulli N. Monitoring of serum enzymes in sport. *Br J Sports Med* 2006; 40: 96–97.
42. de Hoyo M, Cohen DD, Sañudo B, et al. Influence of football match time–motion parameters on recovery time course of muscle damage and jump ability. *J Sports Sci* 2016; 34: 1363–1370.
43. Kim J and Lee J. The relationship of creatine kinase variability with body composition and muscle damage markers following eccentric muscle contractions. *J Exerc Nutrition Biochem* 2015; 19: 123–129.
44. Romagnoli M, Sanchis-Gomar F, Alis R, et al. Changes in muscle damage, inflammation, and fatigue-related parameters in young elite soccer players after a match. *J Sports Med Phys Fitness* 2016; 56: 1198–1205.
45. Bell PG, Stevenson E, Davison GW, et al. The Effects of Montmorency Tart Cherry Concentrate Supplementation on Recovery Following Prolonged, Intermittent Exercise. *Nutrients* 2016; 8: 441.
46. Moore E, Fuller JT, Buckley JD, et al. Impact of cold-water immersion compared with passive recovery following a single bout of strenuous exercise on athletic performance in physically active participants: a systematic review with meta-analysis and meta-regression. *Sports Med* 2022; 52: 1667–1688.

47. Altarriba-Bartes A, Peña J, Vicens-Bordas J, et al. Post-competition recovery strategies in elite male soccer players. Effects on performance: a systematic review and meta-analysis. *PLoS One* 2020; 15: 1–20.
48. Schons P, Preissler AAB, Reichert T, et al. Effects of cold water immersion on the physical performance of soccer players: A systematic review. *Sci Sports* 2022; 37: 159–166.
49. Pooley S, Spendiff O, Allen M, et al. Static stretching does not enhance recovery in elite youth soccer players. *BMJ Open Sport Exerc Med* 2017; 3: e000202.
50. Smith LL, Brunetz MH, Chenier TC, et al. The effects of static and ballistic stretching on delayed onset muscle soreness and creatine kinase. *Res Q Exerc Sport* 1993; 64: 103–107.
51. Dupuy O, Douzi W, Theurot D, et al. An evidence-based approach for choosing post-exercise recovery techniques to reduce markers of muscle damage, {Soreness}, fatigue, and inflammation: {A} systematic review with meta-analysis. *Front Physiol* 2018; 9: 403.
52. Delextrat A, Hippocrate A, Leddington-Wright S, et al. Including stretches to a massage routine improves recovery from official matches in basketball players. *J Strength Cond Res* 2014; 28: 716–727.
53. Hendricks S, Hill H, Hollander Sd, et al. Effects of foam rolling on performance and recovery: a systematic review of the literature to guide practitioners on the use of foam rolling. *J Bodyw Mov Ther* 2020; 24: 151–174.